Experiment - 10

Learning and Assignment of IP Address manually to computer

Aim:

To understand the ip addressing scheme and manually assign static ip address to computers on a network

Procedure:

1) In windows: open network and sharing center -> change Adapter settings -> Right click -> set ip manually

2) Enter IP Address: 192.168.1.10, Subnet mask: 255.255.255.0 Default gateway

3) Enter DNS server: 8.8.8.8 (google DNS) click ok

4) In linux: use command 'sudo ip route addr'

5) To set gateway in linux: 'sudo ip addr add 192.168.1.10/24 dev eth0

commands / code

sudo ip addr add 192.168.1.10/24 dev eth

sudo ip route add default via 192.168.1.

ip a show eth0

Ping 192.168.1.1

nmcli con mod 'wired connection 1' ipv4

nmcli con mod 'wired connection' gateway

nmcli con mod 'wired connection 1' ipv4.meth

nmcli con up 'wired connection 1"

6) For persistent configuration in linux edit /etc/ network /interfaces or uses nmcli

7) verify the configuration using 'ipconfig' or 'ip' a.

8) ping the default gateway to confirm network connectivity

9) ping another computer on the same subnet to verify communication

Result:

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static ip address were successfully assigned to computer. connected was verified using ping commands between hosts on the same subnet